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Synopsis on

"Explainable Data Science"

Submitted in partial fulfilment of the degree of

Master of Computer Applications

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Introduction

Explainable Data Science means making data science and machine learning models easy for people to understand. In many applications, these models help make important decisions such as approving loans, detecting diseases, or recommending products. But some advanced models work like a "black box," where it is hard to know how they made a decision. Explainable Data Science helps explain how these models work and why they give certain results. This makes the system more transparent, trustworthy, and reliable for users.

Objective

To understand what Explainable Data Science means

• To learn why it is important to make data science models easy to understand

• To know how explanations help people trust machine learning systems

• To learn simple methods used to explain how models make decisions

• To understand why transparency and responsibility are important in data-based decisions.

Types of Explainable Data Science

Explainable Data Science is mainly divided into three types:

- **Global Explainability:** Explains how the whole model works and how it makes predictions using different data features.
- **Local Explainability:** Explains why the model made a specific prediction for one particular case.
- **Post-hoc Explainability:** Uses special methods to explain the model's decisions after the model has already been trained, especially for complex models.

Benefits of Explainable Data Science

Explainable Data Science has many advantages:

• Makes machine learning models mor clear and easy to understand

• Helps users trust AI systems more

• Helps find mistakes and bias in models

• Helps people make better decisions

• Ensures the system follows ethical rules and regulations.

Challenges in Explainable Data Science

Some challenges include:

- Advanced machine learning models are very complex and hard to understand
- It is difficult to keep both high accuracy and easy explanation at the same time
- Many people do not fully understand the methods used to explain models
- Large and complex datasets are difficult to manage and analyze.

Applications of Explainable Data Science

Explainable Data Science is used in many areas such as:

- Healthcare systems to help doctors diagnose diseases
- Banking systems to check financial risks and approve loans
- Fraud detection systems to find suspicious activities
- Self-driving cars to understand how decisions are made
- Business systems to help companies make better decisions.

Conclusion

Explainable Data Science is an important part of modern data science and artificial intelligence. It helps users understand how machine learning models make decisions and ensures transparency, fairness, and accountability. By applying explainable techniques, organizations can build more trustworthy and reliable AI systems.